

Preview Engine Design Specification

Ver 0.2

DM270

Department: Imaging

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HISTORY

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1. ABSTRACT

A Preview Engine for DSC27 is a hardware video processor module which generates an image data with YCbCr-4:2:2 format from a CCD raw data with a RGB color space or a complementary color space format. The followings are main features.

- Dual input ports (CCD controller and SDRAM controller)
- Noise filter with noise coring
- Digital gain
- White balance
- Two steps smoothing
- Horizontal and vertical seamless down sampling ($x1/64 \sim x1$)
- Horizontal and vertical seamless zoom ($x1 \sim x4$)*
- RGB2RGB blending matrix
- Gamma correction
- Chroma offset & suppression
- One shot preview
- Support 16bit data bus SDRAM
- Resizing function of YCbCr-4:2:2 image

* Applicable only for SDRAM input mode

Figure 1 shows a block diagram of the preview engine for the DSC27. An input interface module has two input ports which are connected to a CCD controller(CCDC) and a SDRAM controller(SDRAMC), and extracts a valid raw data from a CCD raw data. A noise filter module reduces an impulse noise in the raw data using a noise coring technique. A white balance module has two gain adjusters, a digital gain adjuster and a white balance adjuster. In the digital gain adjuster, the raw data is multiplied by a fixed value of gain regardless of a color space of a CCD image sensor. In the white balance gain adjuster, the raw data is multiplied by an selected value of gain corresponding to the color space of the CCD image sensor. A CFA interpolation module has many sub-modules which processes an interpolation, a smoothing and a resizing. The output from the CFA interpolation module is always formatted into RGB color space regardless of the type of the color space of the CCD. A black adjustment module sets the offset value of RGB data independently to adjust a black level of image data. An RGB2RGB blending module prepares a general 3x3 square matrix and redefines the RGB data from the CFA interpolation module. It can be used as a function of a color correction. A gamma correction module performs a gamma correction to each color by independently prepared gamma ROM table. An RGB2YCbCr conversion module has a 3x3 square matrix circuit and converts the RGB color space of the image data into a YCbCr color space. In a 4:2:2 conversion module, the color data of Cb and Cr are down-sampled into 4:2:2 format. An output interface module has a data buffer for a burst transfer of an SDRAM and requests an arbiter module to store the processed image data. The arbiter module selects one of requests from the input interface of the SDRAMC and the output interface, and requests the SDRAMC to read a source data or to store the processed image data. An ARM interface module is connected to an external ARM bus and send necessary information to each module in the preview engine. The preview engine can be controlled through the ARM bus and the ARM interface.

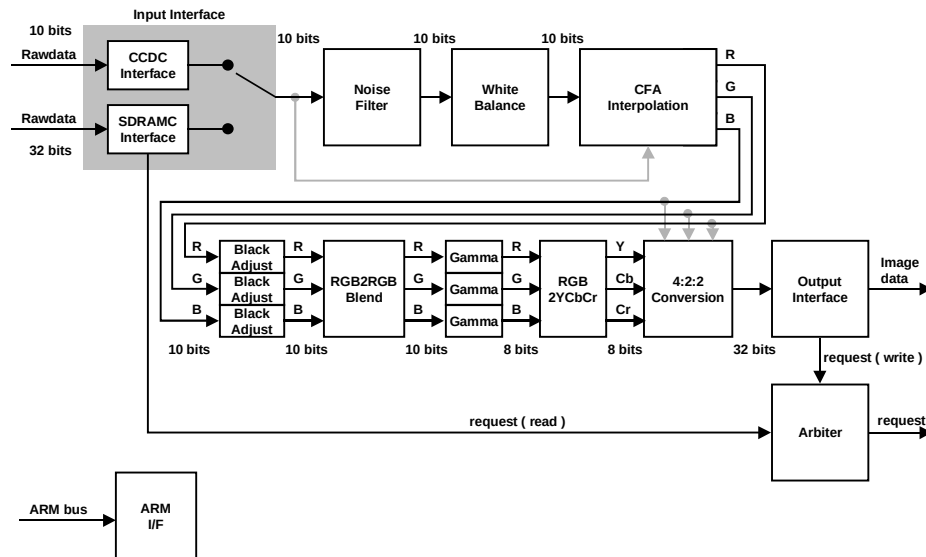


Figure 1 : A Block Diagram of the Preview Engine

2. Port Information

Port Name	Type	Description
vsync_ccd	Input	VSYNC(VD) input from the CCD controller
hsync_ccd	Input	HSYNC(HD) input from the CCD controller
ccdc_wrt	Input	Write enable input from the CCD controller
ccd_data[9..0]	Input	Data input from the CCD controller
sdc_req	Output	Request output to the SDRAM controller
sdc_regen	Input	Request enable input from the SDRAM controller
sdc_rw	Output	Read/Write request output to the SDRAM controller 0 : Write 1 : Read
sdc_16bit	Input	SDRAM data width selection input from the SDRAM controller 0 : 32bit 1 : 16bit
sdc_address[21..0]	Output	SDRAM address output to the SDRAM controller
sdc_data_in[31..0]	Input	SDRAM data input from the SDRAM controller
sdc_data_out[31..0]	Output	SDRAM data output to the SDRAM controller
arm_data_in[15..0]	Input	ARM data input through the ARM bus
arm_add[5..0]	Input	ARM address input through the ARM bus
chip_sel	Input	Chip select input through the ARM bus
arm_write	Input	Write enable input through the ARM bus
arm_data_out[15..0]	Output	ARM data output to the ARM bus
pv_nint	Output	Interrupt output. Issued at the end of the burst transfer to the SDRAMC.
rst_n	Input	Reset input
clk_ccd	Input	CCD clock input
clk_sdr	Input	SDRAM clock input
clk_arm	Input	ARM clock input

Table 2 : port information

3. Operation Guide

Except for selecting a width of SDRAM data bus(32 bits or 16 bits), an activation of the preview engine is determined by registers which are written through the ARM bus(The selection of the width of SDRAM data bus is determined by a special input signal from SDRAMC). In this chapter, some basic operations of the preview engine are explained citing examples. And to simplify an expression, the following two commands “**write**” and “**read**” are used throughout this chapter.

write NN XXXX : write 0XXXXX to a register at offset 0xNN through the ARM bus

read NN : read a value of a register at offset 0xNN through the ARM bus

As to an information of registers, please refer to a register map in chapter 5.

3.1 Start the Preview Engine

By setting a PVEN register at offset 0x00 to 1, the preview engine will be active. And by setting the register to 0, the preview engine will be inactive. Actually, if we set the register to 1, the preview engine becomes active at the next rising edge of the VD signal from the CCDC. And if we set it to 0 under the preview engine is active, the preview engine becomes inactive after the last burst transfer of a processed image data to SDRAMC is finished. At the same time, pv_nint signal is issued synchronized with clk_arm.

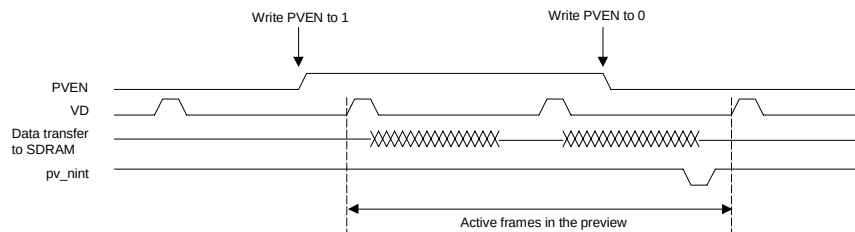


Figure 3.1 : The PVEN register and the activation of the preview engine

The following is an example of an operation to start the preview engine. By reading the PVEN bit, a status of the preview engine can be verified.

```
Ex 3.1) > write 00 0001 // start the preview
        > read 00        // check the PVEN bit
        > 0001           // preview is active
        > write 00 0000 // stop the preview
        > read 00        // check the PVEN bit
        > 0000           // preview is inactive
```

Note) It is recommended that the preview engine is inactive when the following registers is changed.

PVSET1, H_SRT, H_SIZE, V_SRT, V_SIZE, SMTH, H_RSZ, V_RSZ

3.2 Select Input Mode 1

The preview engine has two selectable input ports which are connected to the CCDC and the SDRAMC. Note that there are some difference of available functions between in the CCDC input mode and in the SDRAMC input mode. Table 3.2 shows the main difference of available functions

between the two input modes. As to resize only mode in the SDRAMC input mode, please refer to 3.5.

Functions	CCDC input mode		SDRAMC input mode	
	Normal	Resize Only	Normal	Resize Only
Noise filter	O	X	O	X
Digital gain	O	X	O	X
White balance	O	X	O	X
Smoothing	O	O	O	O
Down sample	O	O	O	O
Zoom	X	X	O	O
RGB blend	O	X	O	X
Gamma	O	X	O	X
Chroma suppress	O	X	O	X
Chroma offset	O	X	O	X
1 shot preview	O	O	O	O

Table 3.2 : Comparison of the two input mode

By setting a INMOD bit in a PVSET1 register at offset 0x01 to 0, the preview engine selects the CCDC input port. And by setting the INMOD bit to 1, the preview engine selects the SDRAMC input port. The following are examples of an operation to select the input mode.

Ex 3.2.1)

```

> write 00 0000 // stop the preview
> write 01 0000 // select the CCDC input mode
> write 04 0000 // set write address 0x000000
> write 05 0000
> write 00 0001 // start the preview

```

Ex 3.2.2)

```

> write 00 0000 // stop the preview
> write 01 0004 // select the SDRAMC input mode
> write 02 0000 // set read address 0x006000
> write 03 6000
> write 04 0000 // set write address 0x000000
> write 05 0000
> write 00 0001 // start the preview

```

3.3 Select input mode 2

The preview engine has three selectable input mode in sdram input mode.

- 10bits mode(8BITS = 0 and ALAW =0 and INMOD = 1 register at offset 0x01)
- 8bits mode (8BITS = 1 and ALAW =0 and INMOD = 1 register at offset 0x01)
- inverse-a-alaw mode(8BITS = 1 and ALAW =1 and INMOD = 1 register at offset 0x01)

3.3.1 10bits mode

The valid data in this mode is 25-16 bit, 9-0 bit in SDRAM data (32bits).

The two data is used from LSM side. Thus the first is 9-0 bit, and the second is 25-16bit.

3.3.2 8bits mode

In this mode the data of SDRAM is used all 32bits and the 32bits data is divided every 8bits. Thus number of data is four in 32bits and the order is from LSB side. And each data is expanded from 8 bits to 10 bits by adding "00" in LSB side.

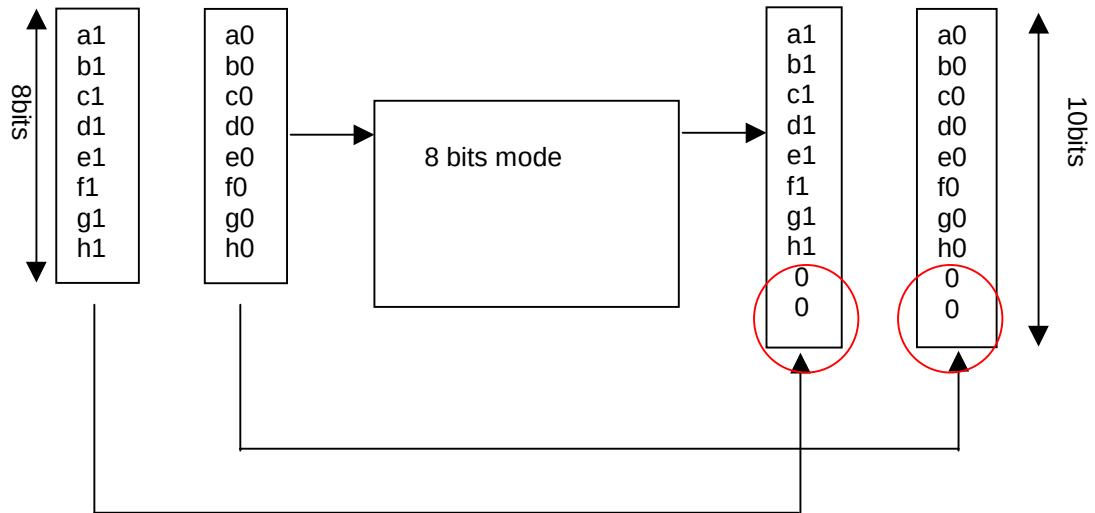


Figure 3.2(a): 8 bits mode

3.3.3 Inverse-a-law mode

In this mode the data of SDRAM is used all 32bits and the 32bits data is divided every 8bits. Thus number of data is four in 32bits and the order is from LSB side. And each data is expanded from 8 bits to 10 bits by using inverse a-law table.

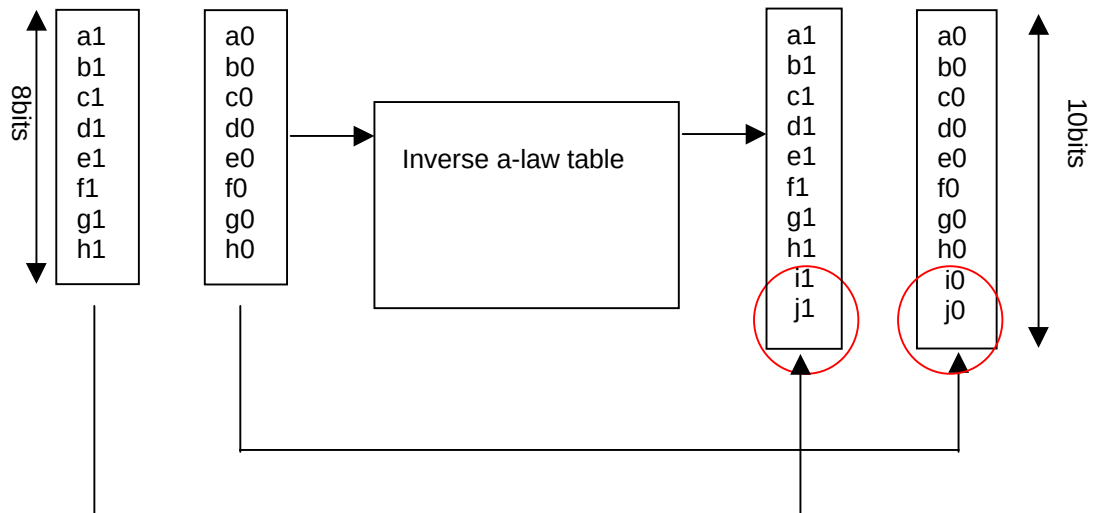


Figure 3.2(b): a law mode

3.4 Define the image size

A size of the output image from the preview engine is determined by the following parameters.

- a valid horizontal image size (a H_SIZE register at offset 0x07)
- a valid vertical image size (a V_SIZE register at offset 0x09)
- a horizontal resize ratio (a H_RSZ register at offset 0x1E)
- a vertical resize ratio (a V_RSZ register at offset 0x1F)

And each size of the output image, H_{out} and V_{out} , is expressed by the following equations.

$$H_{out} = \begin{cases} H_SIZE \times \frac{16}{H_RSZ} & (\text{if } H_RSZ > 0x0010) \\ (H_SIZE - 1) \times \frac{16}{H_RSZ} & (\text{if } H_RSZ < 0x0010) \end{cases}$$

$$V_{out} = V_SIZE \times \frac{16}{V_RSZ}$$

The horizontal size of the output image must be lower than 720 pixels due to the size of implemented line memories. For example, when we apply a x4 zoom and want the size of output image to be

$$720 = (H_SIZE - 1) \times \frac{16}{0x0004}$$

$$\therefore H_SIZE = 181 = 0x00B5$$

$$480 = V_SIZE \times \frac{16}{0x0004}$$

$$\therefore V_SIZE = 120 = 0x0078$$

720x480, that is, $H_RSZ=0x0004$, $V_RSZ=0x0004$, $H_{out}=720$ and $V_{out}=480$, an appropriate H_SIZE and V_SIZE can be calculated as follows.

The following is an example operation to apply this x4 zoom function.

```
Ex 3.3) > write 00 0000 // stop the preview
> write 01 0004 // SDRAMC input mode
> write 07 00B5 // Set a valid H-size
> write 09 0078 // Set a valid V-size
> write 1E 0004 // Set a H-resize ratio
> write 1F 0004 // Set a V-resize ratio
> write 00 0001 // start the preview
```

Note) Depending on the values of the H_RSZ and the V_RSZ, the preview engine sends the excessive image data to the SDRAM. Therefore, please prepare at least more one burst of the excessive SDRAM work area than that for the image size to be displayed.

3.5 One shot preview

By setting a PVOS bit in the PVSET1 register at offset 0x01 to 1, the preview engine processes only one frame of a CCD raw data. When the preview engine finishes the processing and transferring the image data to the SDRAMC, the preview engine stops by itself.

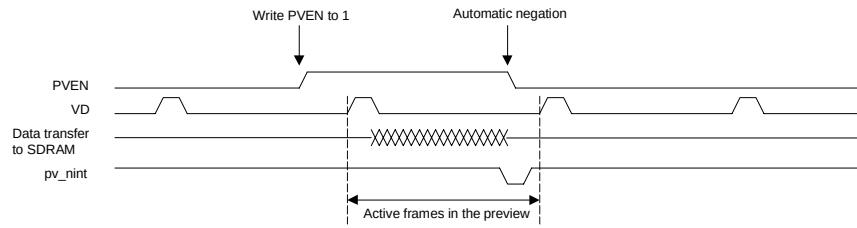


Figure 3.4 : Activation of preview engine in one shot preview mode

The following is an example to perform the one shot preview mode and check a status of the PVEN bit.

```
Ex 3.4) > write 00 0000 // stop the preview
> write 01 0010 // 1 shot preview mode
           // CCDC input mode
> write 00 0001 // start the preview
> read 00      // read PVEN register
> 0000        // preview is stopped automatically
```

3.6 Resizing function for YCbCr image data

The preview engine has another data path to utilize only a resizing function in the CFA interpolation module. Therefore, we can use the preview engine as an only resizing function to image data with YCbCr-4:2:2 format. Figure 3.5 shows the data path of the preview engine in this “resizing only mode”.

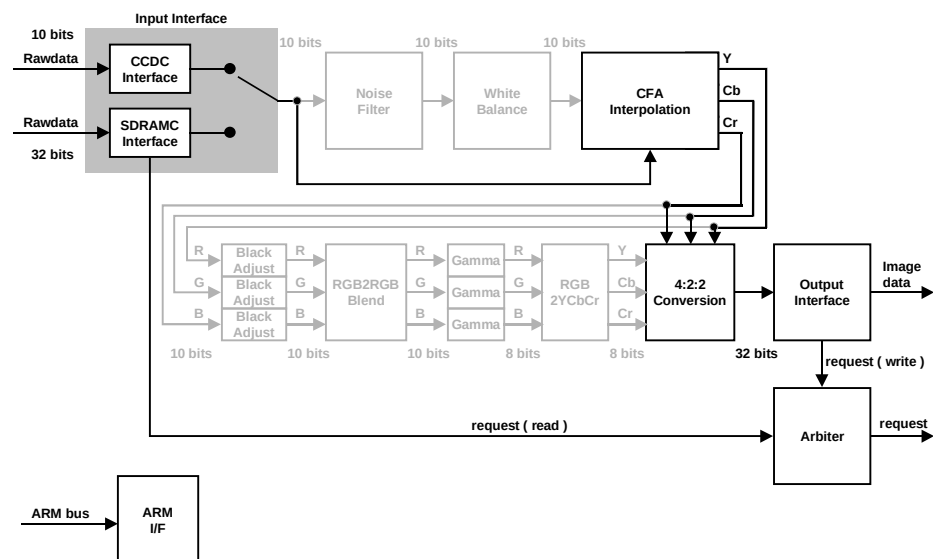
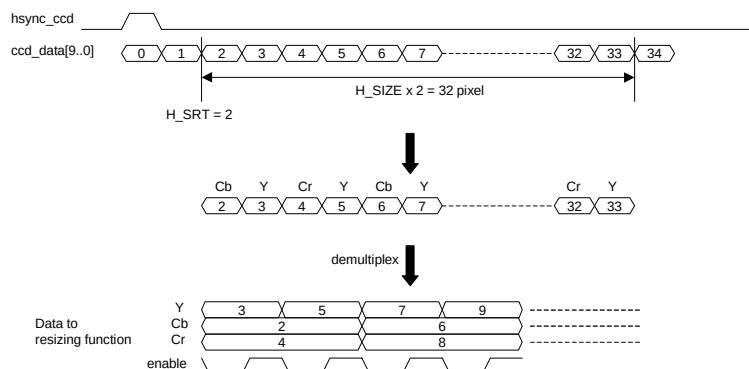


Figure 3.5(a) : The data path in resizing only mode

In the CCDC input mode, note that the preview engine considers the sequence of input data from CCDC to CbYCrYCbYCrY.... according to a H_SRT register at offset 0x06. Figure 3.5(b) and Example 3.5 show the example of this resizing only mode in the CCDC input mode.



**Figure 3.5(b) : Example of the resizing only mode in the CCDC input mode
(H_SRT=0x0002, H_SIZE=0x0020)**

```

Ex 3.5) > write 00 0000 // stop the preview
> write 01 0008 // resample only mode
           // CCDC input mode
> write 06 0002 // set horizontal start point 0x0002
> write 07 0010 // set horizontal size 0x0010 (16 pixel)
> write 00 0001 // start the preview
  
```

In the SDRAM input mode, size of the resized image can be calculated samely as in the normal SDRAM input mode described in 3.3.

4. Functional Description

4.1 Noise Filter

A noise filter is the first image processing in the preview engine. It is useful to reduce an impulse noise included in the raw data from the CCD image sensor using a noise coring technique. When an original pixel to be applied is set to X_0 , the following calculation is made.

$$x_0 = X_0 - a \times \frac{X_{-2} - 2X_0 + X_2}{4}$$

$$(a = \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1)$$

Figure 4.1 shows the block diagram of the noise filter module. It is composed of a high pass filter, a shifter, a subtracter and a output selector. When a NF_EN bit in a N_FIL register is 1, this noise filter is valid. And NF_RT[1..0] bits in the N_FIL register controls the gain of the high pass filter. As to the N_FIL register, please refer to 5.12.

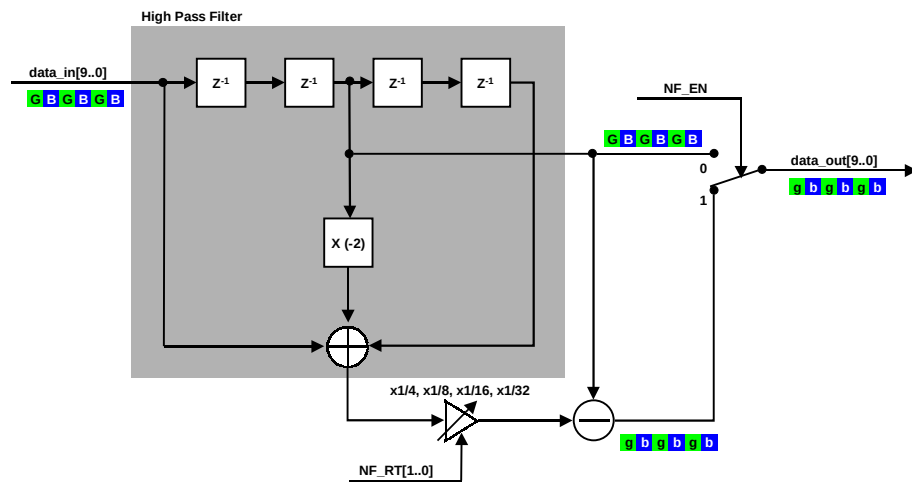


Figure 4.1 : Block diagram of the noise filter module

4.2 White Balance

A white balance module has two gain adjusters, a digital gain adjuster and a white balance adjuster. In the digital gain adjuster, the raw data is multiplied by a fixed value gain regardless of a color pixel to be processed. In the white balance gain adjuster, the raw data is multiplied by a selected gain corresponding to the color of the processed pixel. Figure 4.2(a) shows this example.

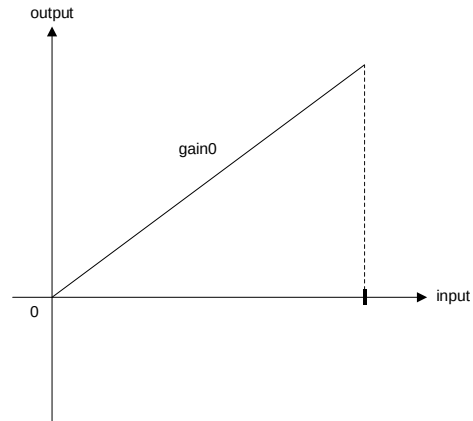


Figure 4.2(a) : Example of the white balance

Figure 4.2(b) shows the block diagram of the white balance module. It is composed of two multipliers for the digital gain and the white balance gain.

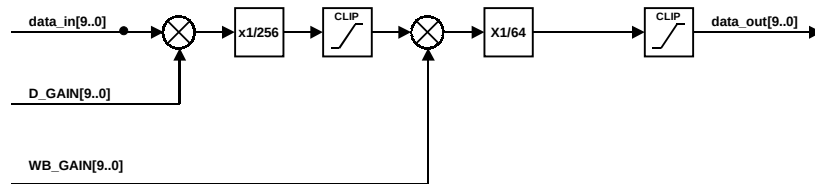


Figure 4.2(b) : Block diagram of the white balance module

4.3 CFA Interpolation

In a CFA interpolation module, many important sub-modules for an image processing are included such as a horizontal and vertical interpolation, a smoothing and a horizontal and vertical resizing. Figure 4.3(a) and (b) show the block diagram and the data path of the CFA interpolation module in the normal mode and the resize only mode. A horizontal interpolation sub-module processes a horizontal interpolation using a two tap or a five tap interpolation filter. A smoother sub-module performs a selectable low pass filter which has 2 different levels of a band-width. A horizontal resampler sub-module performs a seamless horizontal down-sampling or up-sampling (zoom) by x4 over-sampling. Similarly, a vertical interpolation & resampler sub-module performs a vertical interpolation and a seamless vertical down-sampling or up-sampling by x4 over-sampling using two line memories. A color selector sub-module extracts an appropriate color from the 4 input from the vertical interpolation and resampler sub-module. A color adjustment sub-module executes the color adjustment to an interpolated data of 4 colors in the complementary color space. In case of the RGB CCD mode, the data bypass this sub-module. A comp2RGB conversion sub-module converts the data from 4 color in the complementary color space to the RGB color space. Therefore, the output of

this CFA interpolation module is formatted to RGB color space regardless of a type of the color space of CCD image sensor. As shown in Figure 4.3(b), in the resample only mode, only the smoother, the horizontal and vertical resampler are processed.

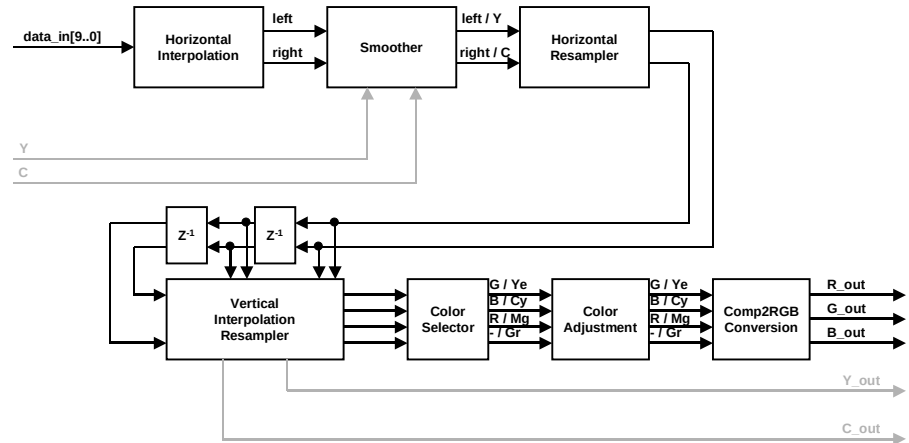


Figure 4.3(a) : Block diagram of the CFA interpolation module (Normal mode)

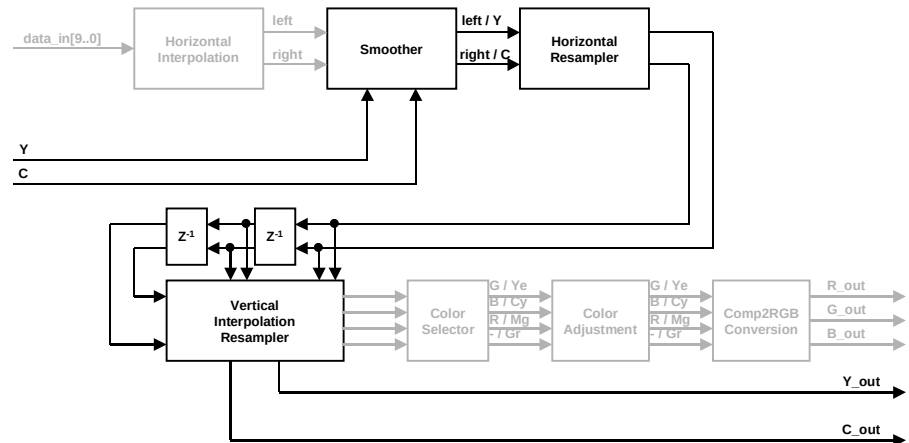


Figure 4.3(b) : Block diagram of the CFA interpolation module (Resize only mode)

4.3.1 Horizontal Interpolation

A horizontal interpolation sub-module performs a horizontal interpolation with a two tap or a five tap interpolation filter. The 2 taps interpolation filter uses a simple average of the next pixels on the left and right side of the pixel to be interpolated. On the other hand, the 5 taps interpolation filter utilizes the information of another color on the processed line so that a false color due to this interpolation process can be reduced effectively. Actually, when a pixel at the position to be interpolated is set to X_0 , this horizontal interpolation sub-module performs the following calculation.

$$x_0 = \begin{cases} \frac{X_{-1} + X_1}{2} & (2 \text{ taps}) \\ \frac{-X_{-2} + 2X_{-1} + 2X_0 + 2X_1 - X_2}{4} & (5 \text{ taps}) \end{cases}$$

Figure 4.3.1 shows a block diagram of the horizontal interpolation sub-module. It is composed three adders for calculating the above expression. A type of the two interpolation processes depends on a H_INTP bit in PVSET2 register. Please refer to 5.11.

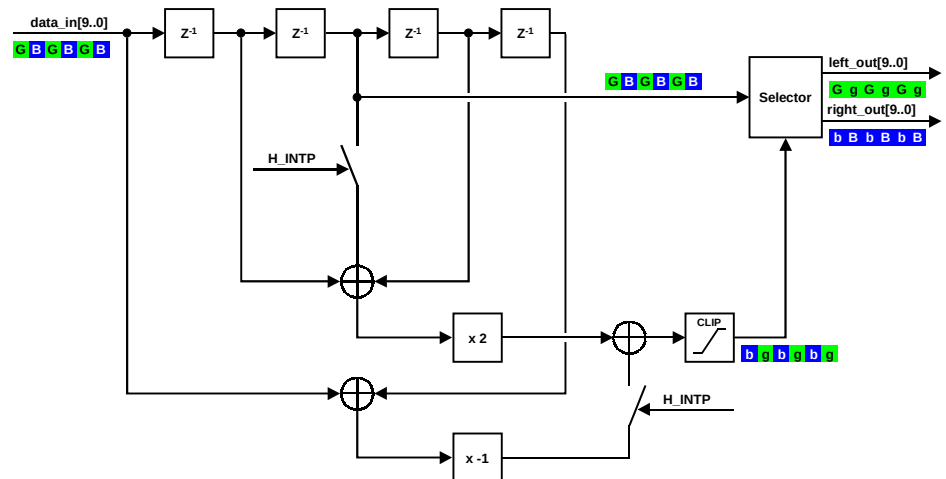


Figure 4.3.1 : Block diagram of the horizontal interpolation sub-module

4.3.2 Smoother

A smoother sub-module is an anti-aliasing filter which is composed of two kinds of FIR filters. By combining the two filter, one of two levels of horizontal low pass filter can be performed. Actually, one of the following calculation is made depending on the register setting.

$$x_0 = \begin{cases} X_0 & \text{(bypass)} \\ \frac{X_{-1} + 2X_0 + X_1}{4} \equiv x'_0 & \text{(weak)} \\ \frac{x'_{-2} + 2x'_0 + x'_2}{4} & \text{(strong)} \end{cases}$$

Figure 4.3.2 shows a block diagram of the smoother sub-module. It is composed of the two FIR filter and an output selector. The output is determined by a SMTH_EN bit and a SMTH_LV bit in a SMTH register. Please refer to 5.30.

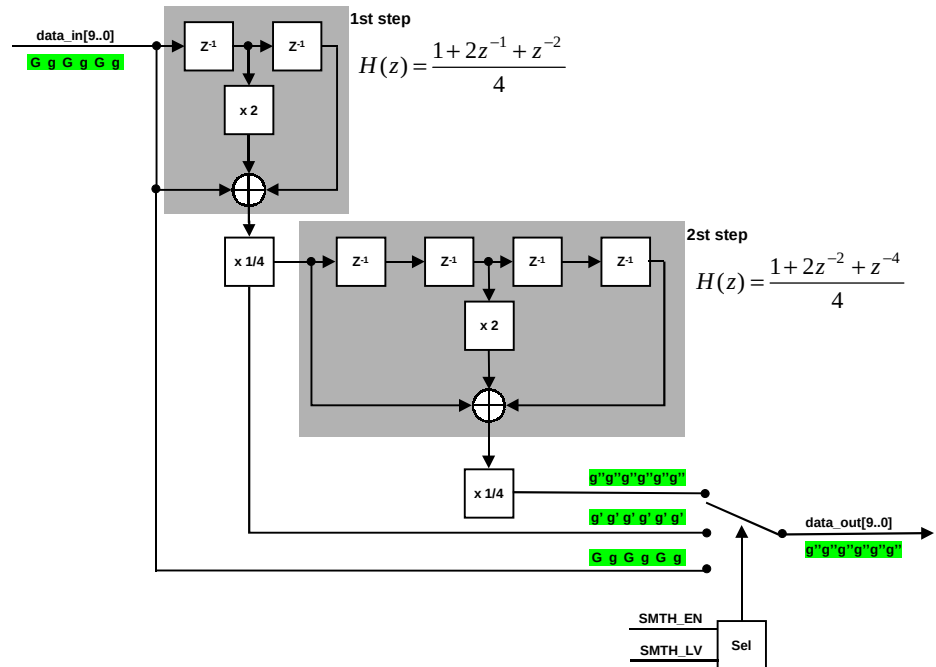


Figure 4.3.2 : Block diagram of the smoother sub-module

4.3.3 Horizontal Resampler

A horizontal resampler sub-module performs a seamless horizontal down-sampling or up-sampling (zoom). This seamless resampling processing is done by a bi-linear interpolation of data from the smoother sub-module. Figure 4.3.3(a) shows an example of this resampling process (x2 zoom).

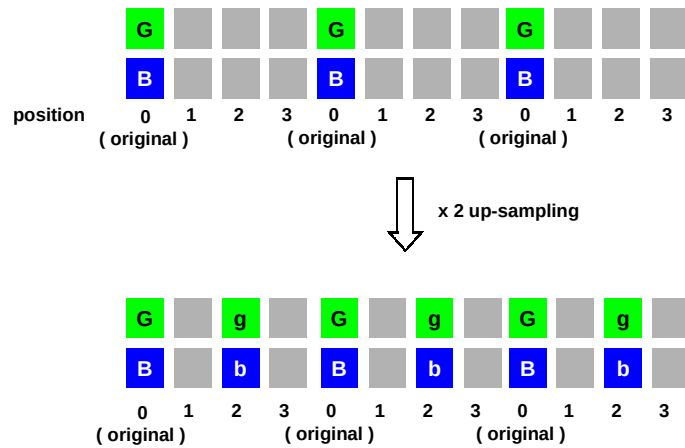


Figure 4.3.3(a) : Example of the resampling (x2 up-sampling)

Actually, one of the following four calculation are processed depending on a resampled position.

$$x_0 = \begin{cases} X_0 & (\text{position } 0) \\ \frac{X_0 + 3X_1}{4} & (\text{position } 1) \\ \frac{X_0 + X_1}{2} & (\text{position } 2) \\ \frac{3X_0 + X_1}{4} & (\text{position } 3) \end{cases}$$

Figure 4.3.3(b) shows a block diagram of the horizontal resampler sub-module. The calculation of the resampled position above is done by a position detector which accumulates H_RSZ bits in a H_RSZ register periodically.

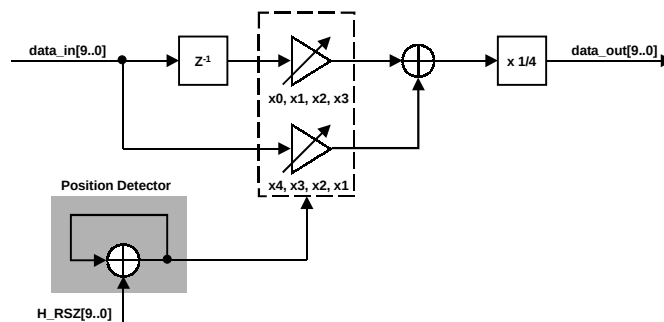


Figure 4.3.3(b) : Block diagram of the horizontal resampler sub-module

4.3.4 Vertical Interpolation & Resampler

A vertical interpolation & resampler sub-module performs a vertical interpolation and a seamless vertical down-sampling or up-sampling. Same as the horizontal resampler module, this vertical resampling is processed by a bi-linear interpolation of a data from the horizontal interpolation sub-module and two line memories. And this sub-module executes two different calculation depending on the color pattern of CCD image sensor. Figure 4.3.4(a) shows an example of this two types of the CFA pattern.

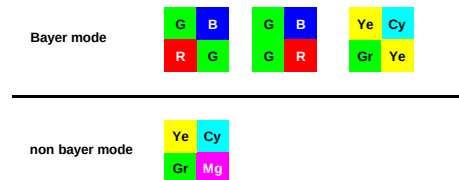


Figure 4.3.4(a) : Types of CFA pattern

If there is a color which exists in every line, this sub-module activates in “bayer mode”, otherwise it activates in “non-bayer mode”. In the bayer mode, the vertical interpolation and resampling processing are based on the information of a main color (G or Ye in Figure 4.3.4(a)). As shown in Figure 4.3.4(b), in the bayer mode, one main color (G in this figure) exists in every line. Therefore, in this mode, the other colors can be calculated in order to keep relationship with the main color. Actually in this example of the bayer mode, the following calculation is made in this sub-module.

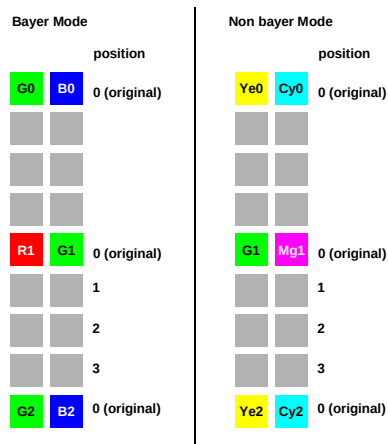


Figure 4.3.4(b) : Comparison of inputs in bayer mode and in non bayer mode

Position 0

$$r = R_1$$

$$g = G_1$$

$$b = \frac{B_0 + B_2}{2} - \frac{G_0 + G_2}{2} + G_1$$

Position 2

$$r = R_1 - G_1 + g$$

$$g = \frac{G_1 + G_2}{2}$$

$$b = \frac{B_0 + 3B_2}{4} - \frac{G_0 + 3G_2}{4} + g$$

Position 1

$$r = R_1 - G_1 + g$$

$$g = \frac{3G_1 + G_2}{4}$$

$$b = \frac{3B_0 + 5B_2}{8} - \frac{3G_0 + 5G_2}{8} + g$$

Position 3

$$r = R_1 - G_1 + g$$

$$g = \frac{G_1 + 3G_2}{4}$$

$$b = \frac{B_0 + 7B_2}{8} - \frac{G_0 + 7G_2}{8} + g$$

On the other hand, in the non bayer mode, there is no main color in the CFA pattern. Therefore, the vertical interpolation and resampling processing is performed in order to keep relationship with upper or lower color (in Figure 4.3.4(b), Ye and G, Cy and Mg). Actually, the following calculation are made.

Position 0

$$ye = \frac{Ye_0 + Ye_2}{2}$$

$$cy = \frac{Cy_0 + Cy_2}{2}$$

$$gr = Gr_1$$

$$mg = Mg_1$$

Position 1

$$ye = \frac{3Ye_0 + 5Ye_2}{8}$$

$$cy = \frac{3Cy_0 + 5Cy_2}{8}$$

$$gr = Gr_1 - \frac{Ye_0 + Ye_2}{2} + ye$$

$$mg = Mg_1 - \frac{Cy_0 + Cy_2}{2} + cy$$

Position 2

$$ye = \frac{Ye_0 + 3Ye_2}{4}$$

$$cy = \frac{Cy_0 + 3Cy_2}{4}$$

$$gr = Gr_1 - \frac{Ye_0 + Ye_2}{2} + ye$$

$$mg = Mg_1 - \frac{Cy_0 + Cy_2}{2} + cy$$

Position 3

$$ye = \frac{Ye_0 + 7Ye_2}{8}$$

$$cy = \frac{Cy_0 + 7Cy_2}{8}$$

$$gr = Gr_1 - \frac{Ye_0 + Ye_2}{2} + ye$$

$$mg = Mg_1 - \frac{Cy_0 + Cy_2}{2} + cy$$

Figure 4.3.4(c), (d) and (e) show a block diagram of the vertical interpolation and resampler sub-module. There are two steps of the processing for supporting the two types of calculation in the same circuit. The first step is to prepare all terms of expression above calculating the resampled position as in the horizontal resampling sub-module. In the second step, the final outputs are calculated using the results of the first step operation.

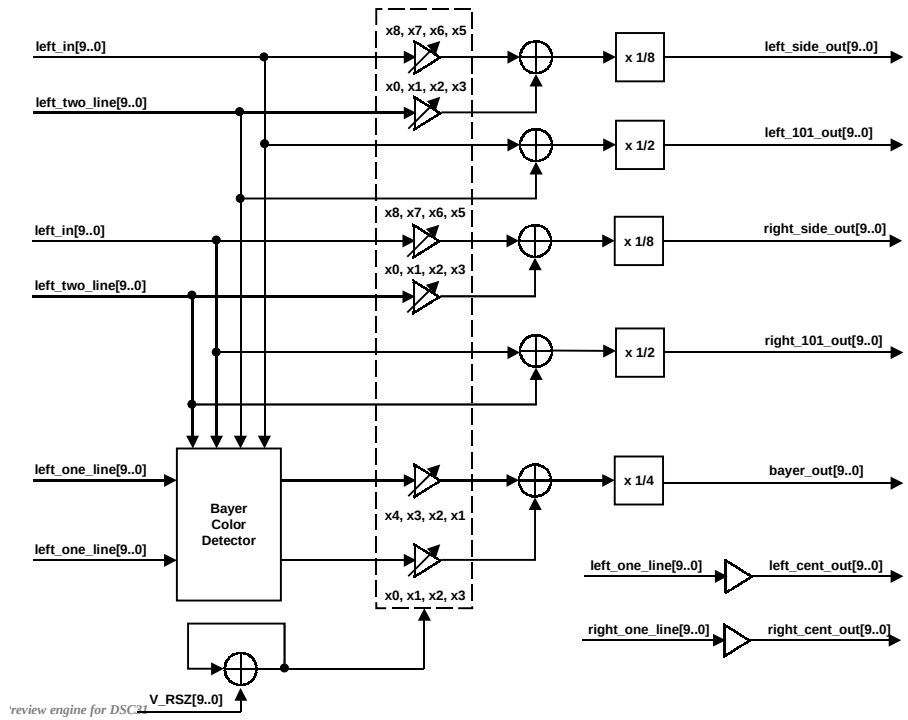


Figure 4.3.4(c) : The 1st step in the vertical interpolation sub-module

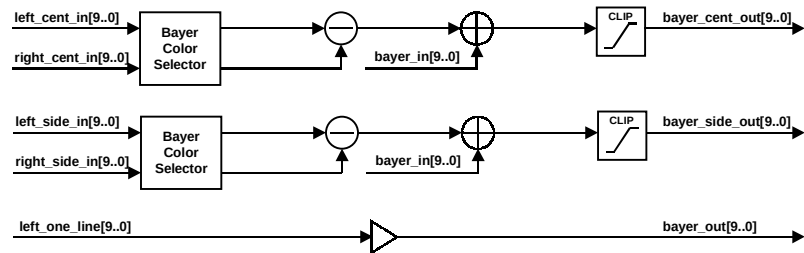


Figure 4.3.4(d) : The 2nd step in the vertical interpolation sub-module (bayer mode)

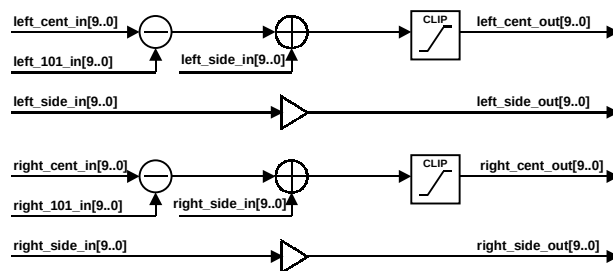


Figure 4.3.4(e) : The 2nd step in the vertical interpolation sub-module (non bayer mode)

4.3.5 Color Selector

A color selector sub-module extracts an appropriate color from the four input signals which the vertical interpolation and resampler sub-module generates.

Figure 4.3.5 shows a block diagram of the color selector sub-module. These extractor for each color select an appropriate signal automatically using the sync signal and CFA[7..0] bits in the PVSET2 register. Please see also 5.11.

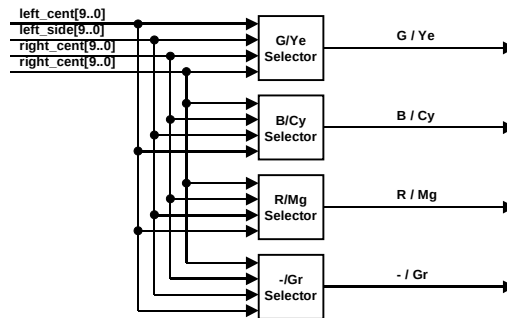


Figure 4.3.5 : Block diagram of the color selector sub-module

4.3.6 Color Adjustment for the complementary color space

A color adjustment sub-module performs the adjustment to interpolated data of 4 colors in the complementary color space in order to satisfy an equation of the following complementary color space.

$$Ye + Cy = Gr + Mg$$

To keep this relationship, firstly, the following value e is calculated.

$$e = G + Mg - Ye - Cy$$

This value e can be defined as the amount of errors of 4 colors from the interpolation process. Therefore, the result from the interpolation process, Ye , Cy , Gr and Mg , is redefined to satisfy the above formula.

$$\begin{cases} ye = Ye + \frac{e}{4} \\ cy = Cy + \frac{e}{4} \\ gr = Gr - \frac{e}{4} \\ mg = Mg - \frac{e}{4} \end{cases}$$

Figure 4.3.6 shows a block diagram of the color adjustment sub-module. It is composed of six adders and one subtracter for executing the above calculation. If a CCDMOD bit in the PVSET2 register is zero, the data bypass this sub-module.

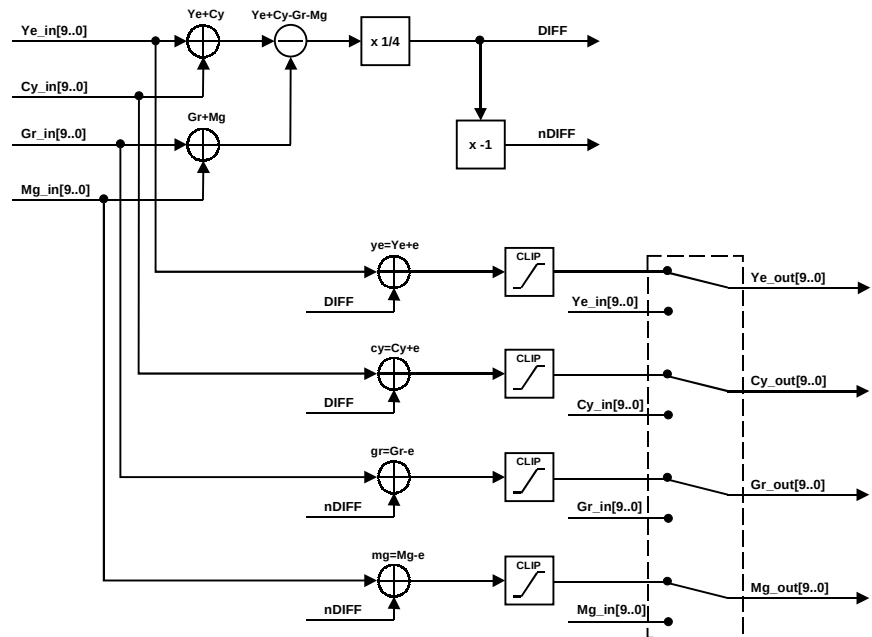


Figure 4.3.6 : Block diagram of the color adjustment sub-module

4.3.7 Comp2RGB conversion

A comp2RGB conversion sub-module converts the data from four colors in the complementary color space to the RGB color space. Especially for G, a data from the interpolation process and data calculated in this module using the same formula in the complementary color space can be blended. Actually, the following calculation is made based on the conversion formula.

$$\begin{cases} r = Ye - Cy + Mg \\ g = rG + (1-r)(Ye + Cy - Mg) \quad (r = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1) \\ b = Mg - Ye + Cy \end{cases}$$

Figure 4.3.7 shows a block diagram of the color adjustment sub-module. Same as in the color adjustment sub-module, if the CCDMOD bit is 0, the data bypass this module.

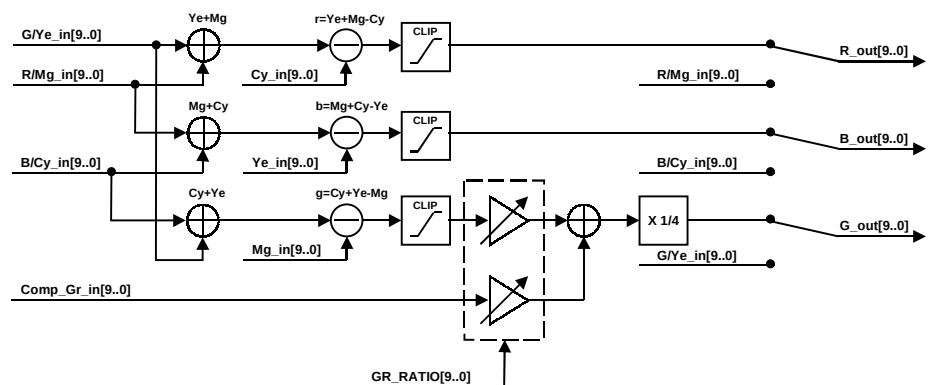


Figure 4.3.7 : Block diagram of Comp2RGB conversion

4.4 Black Adjuster Module

A black adjuster module performs the following calculation for an adjustment of each color level.

$$data_out = data_in + bl_offset$$

Figure 4.4 shows the block diagram of this black adjuster module. A simple one addition and clip operation are processed in this module. The output $data_out[10..0]$ is a 2's complement format.

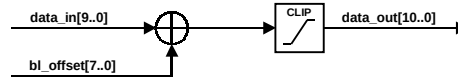


Figure 4.4 : Block diagram of black adjuster module

4.5 RGB2RGB Blending Matrix

An RGB2RGB blending module has a general 3x3 square matrix and redefines the RGB data from the CFA interpolation module, which can be used as a function of a color correction. In this module, the following calculation is made.

$$\begin{bmatrix} R_{out} \\ G_{out} \\ B_{out} \end{bmatrix} = \begin{bmatrix} MTX_{RR} & MTX_{GR} & MTX_{BR} \\ MTX_{RG} & MTX_{GG} & MTX_{BG} \\ MTX_{RB} & MTX_{GB} & MTX_{BB} \end{bmatrix} \begin{bmatrix} R_{in} \\ G_{in} \\ B_{in} \end{bmatrix} + \begin{bmatrix} R_offset \\ G_offset \\ B_offset \end{bmatrix}$$

Figure 4.5 shows a block diagram of the RGB2RGB blending module. Nine multipliers and six adders are required for performing this matrix operation.

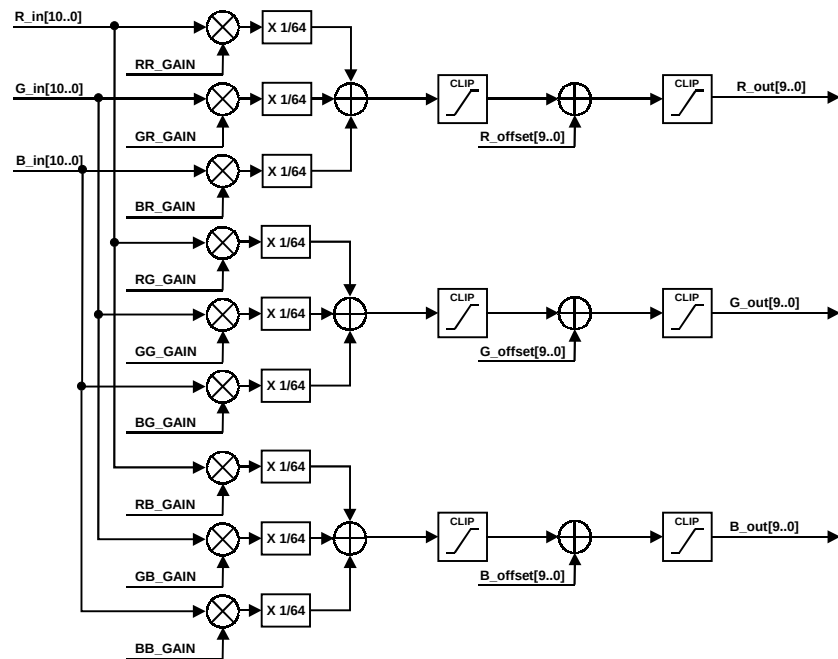


Figure 4.5 : Block diagram of the RGB2RGB blend module

4.6 Gamma Correction

A gamma correction module performs a gamma correction independently for each color in the RGB color space by the specific ROMs. As shown in Figure 1, this module exists independently for each color so that the independent setting is possible.

Figure 4.6 shows a block diagram of the gamma correction module. It is composed of one ROM table and one selector. When the BYPASS bit is asserted, the input data is only divided by four.

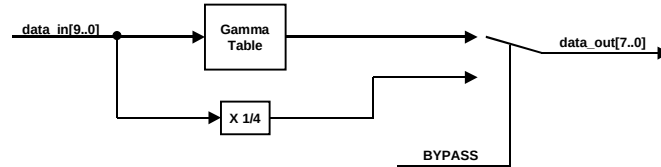


Figure 4.6 : Block diagram of the gamma correction module

4.7 RGB2YCbCr Conversion Matrix

An RGB2YCbCr conversion module has a 3x3 square matrix and converts the RGB color space of the image data into a YCbCr color space. In addition to the conversion matrix operation, the functions of an offset, a contrast, a brightness and a chroma suppression are performed in this module.

Figure 4.7 shows a block diagram of the RGB2YCbCr conversion matrix. It is composed of nine multipliers and three adders for the basic conversion matrix, two multipliers for the chroma suppression and 2 adders for the chroma offset.

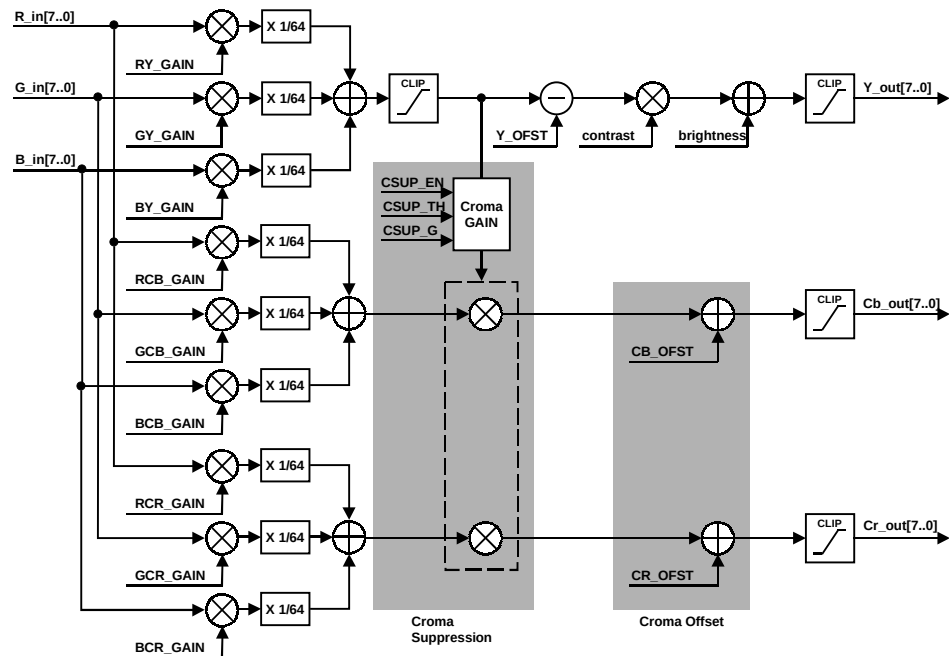


Figure 4.7 : Block diagram of the RGB2YCbCr conversion module

5. Register Map

PVEN (0003:0780) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PVEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved Bit. Do not use.
0	PVEN	Preview Engine enable. 0: Disable 1: Enable

PVSET1 (0003:0782) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	8BITS	ALAW	PVOS	RSOLY	INMOD	BSTAL	WEN
R/W	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0

Bits	Register Name	Description
15:7	RSV	Reserved Bit. Do not use.
6	8BITS	8BITS mode 0: 10 bits mode 1: 8 bits mode
5	ALAW	Inverse A-LAW mode 0: Normal mode 1: inverse A-LAW mode (and 8BITS = 1)
4	PVOS	One shot preview 0: Normal (continuous process) 1: One shot (Outputs only one screen)
3	RSOLY	Resize only. Selecting the process contents. 0: Normal process. The input data is Raw data 1: Resize process only. The input data is YcbCr 4:2:2
2	INMOD	Input Mode. Select the Input signal. 0: Input from CCDC 1: Input from SDRAM
1	BSTAL	Burst align 0: Normal forwarding/Transferring 1: Forwarding /Transferring by burst boundary size.
0	WEN	Write enable. Set if the input write enable from CCDC is Valid or Invalid. 0: Invalid 1: Valid

RADDU (0003:0784) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RADD2	RADD2	RADD1	RADD1	RADD1	RADD1
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:6	RSV	Reserved Bit. Do not use.
5:0	RADD5-0	When data from SDRAM is read, specify the upper/above address of first address

RADDL (0003:0786) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RADD15	RADD14	RADD13	RADD12	RADD11	RADD10	RADD9	RADD8	RADD7	RADD6	RADD5	RADD4	RADD3	RADD2	RADD1	RADD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	RADD15-0	When data from SDRAM is read, specify the lower/below address of first address

WADU (0003:0788) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	WAD21	WAD20	WAD19	WAD18	WAD17	WAD16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:6	RSV	Reserved Bit. Do not use.
5:0	WADD5-0	When data from SDRAM is write, specify the upper/above address of first address

WADL (0003:078A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WAD15	WAD14	WAD13	WAD12	WAD11	WAD10	WAD9	WAD8	WAD7	WAD6	WAD5	WAD4	WAD3	WAD2	WAD1	WAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	RADD15-0	When data from SDRAM is write, specify the lower/below address of first address

HSRT (0003:078C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HSRT11	HSRT10	HSRT9	HSRT8	HSRT7	HSRT6	HSRT5	HSRT4	HSRT3	HSRT2	HSRT1	HSRT0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved Bit. Do not use.
11:0	HRST11-0	Horizontal start position. When INMOD=0 , the beginning position of taking data to SDRAM is set by number of pixel from HD. When INMOD=1, the horizontal size of the data read from SDRAM is set by the number of pixels.

HSIZE (0003:078E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	HSIZE11	HSIZE10	HSIZE9	HSIZE8	HSIZE7	HSIZE6	HSIZE5	HSIZE4	HSIZE3	HSIZE2	HSIZE1	HSIZE0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved Bit. Do not use.
11:0	HSIZE11-0	Horizontal Size The horizontal size of taking data to SDRAM is set by the number of pixels. When INMOD=0, It is necessary to set more than 32 pixels from the number of horizontal total pixels by a small value. In other words, the Blanking period should be 32 or more.

VSRT (0003:0790) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VSRT11	VSRT10	VSRT9	VSRT8	VSRT7	VSRT6	VSRT5	VSRT4	VSRT3	VSRT2	VSRT1	VSRT0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved Bit. Do not use.
11:0	VRST11-0	Vertical start position. The beginning position of taking data to SDRAM is set by number of lines from VD.

VSIZ (0003:0792) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	VSIZ11	VSIZ10	VSIZ9	VSIZ8	VSIZ7	VSIZ6	VSIZ5	VSIZ4	VSIZ3	VSIZ2	VSIZ1	VSIZ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserved Bit. Do not use.
11:0	VSIZ11-0	Vertical Size A vertical size of taking data to SDRAM is set by the number of lines.

PVSET2 (0003:0794) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	H_INTP	V_INTP	NFIL	CCDMD	RSV	GBLD2	GBLD1	GBLD0	CFA7	CFA6	CFA5	CFA4	CFA3	CFA2	CFA1	CFA0
R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	H_INTP	Seelction of Horizontal process. 0: Other peripheral color information is used (5 type). 1: Only the same color is used (2 type).
14	V_INTP	Seelction of vertical process. 0: Other peripheral color information is used (5 type). 1: Only the same color is used (2 type).

Bits	Register Name	Description
13	V_NFIL	Vertical noise filter 0: Off 1: On
12	CCDMD	CCDMode 0: RGB 1: Complementary color
11	RSV	Reserved Bit. Do not use.
10:8	GBLD2-0	Selection of G process. When complementary colour CCD is used, the G signal Process is selected. In $G_{out} = rG_{org} + (1-r)(Y_e + C_y - M_g)$ GBLD2 GBLD1 GBLD0 0 0 0 $r=1$ 0 0 1 $r=3/4$ 0 1 0 $r=1/2$ 0 1 1 $r=1/4$ 1 0 0 $r=0$
7:0	CFA7-0	Specification of coloursThe color of the pixel is set according to the color filter of CCD. CFA7-6: Even number pixel/Even number lines CFA5-4: Odd number Pixel/Even number lines CFA3-2: Even number pixel/Odd number lines CFA1-0: Odd number Pixels/Odd number lines CFA7,5,3,1 CFA6,4,2,0 0 0: G/Ye 0 1: B/Cy 1 0: R/Mg 1 1: -/G

N_FIL (0003:0796) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	NF_EN	RSV	RSV	NF_RT1	NF_RT0
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	0	0

Bits	Register Name	Description
15:5 3:2	RSV	Reserved Bit. Do not use.
4	NF_EN	The horizontal noise filter 0: Off 1: On
1:0	NF_RT1-0	Setting the horizontal noise filter level NF_RT1 NF_RT0 Noise removal level 0 0 Less 0 1 1 0 1 1 More

DGAIN (0003:0798) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	DGAIN9	DGAIN8	DGAIN7	DGAIN6	DGAIN5	DGAIN4	DGAIN3	DGAIN2	DGAIN1	DGAIN0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-10	RSV	Reserved Bit. Do not use.
9-0	DGAIN9-0	Digital gain. It is possible to set within the range of $x0 \sim x3.99609375(3+255 / 256)$. The decimal point is located between DGAIN8 and DGAIN7. DGAIN9 - DGAIN0 11 1111 1111: $x3.99609375(3 + 255 / 256)$ 01 0000 0000: $x1$ 00 1000 0000: $x0.5$ 00 0000 0001: $x0.00390625(1/256)$ 00 0000 0000: $x0$

WB_GAIN0 (0003:079A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WG007	WG006	WG005	WG004	WG003	WG002	WG001	WG000	WG017	WG016	WG015	WG014	WG013	WG012	WG011	WG010
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	WG007-0	White balance gain setting. It is possible to set within the range of area 0 and Gb/Ye pixel $x0 \sim x7.96875(7+31 / 32)$. The decimal point is located between WG006 and WG005.
7:0	WG017-0	White balance gain setting. B/Cy pixel

WB_GAIN1 (0003:079C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WG027	WG026	WG025	WG024	WG023	WG022	WG021	WG020	WG037	WG036	WG035	WG034	WG033	WG032	WG031	WG030
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	WG027-0	White balance gain setting. Gr/G pixel
7:0	WG037-0	White balance gain setting. R/Mg pixel

WGxx7 - WGxx0	Gain
1111 1111	$x3.984375(3 + 63/64)$
0100 0000	$x1$
0010 0000	$x0.5$
0000 0001	$x0.03125(1/64)$
0000 0000	$x0$

SMTH (0003:079E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SMTEN	SMTLV
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
15:2	RSV	Reserved Bit. Do not use.
1	SMTEN	Smoothing enable 0: Disable 1: Enable
0	SMTLV	Smoothing level 0: Less 1: More

H_RSZ (0003:07A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	HRSZ9	HRSZ8	HRSZ7	HRSZ6	HRSZ5	HRSZ4	HRSZ3	HRSZ2	HRSZ1	HRSZ0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved Bit. Do not use.
9:0	HRSZ9-0	Horizontal Resizing. When horizontal resizing is done, the denominator of the below mentioned formula is set. RH shows the ratio of resizing. $R_H = \frac{16}{HRS}$

V_RSZ (0003:07A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	VRSZ9	VRSZ8	VRSZ7	VRSZ6	VRSZ5	VRSZ4	VRSZ3	VRSZ2	VRSZ1	VRSZ0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved Bit. Do not use.
9:0	VRSZ9-0	Vertical resizing. When vertical resizing is done, the denominator of the below mentioned formula is set. R _V shows the ratio of resizing. $R_V = \frac{16}{VRS}$

BL_OFST0 (0003:07A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BL_R7	BL_R6	BL_R5	BL_R4	BL_R3	BL_R2	BL_R1	BL_R0	BL_G7	BL_G6	BL_G5	BL_G4	BL_G3	BL_G2	BL_G1	BL_G0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	BL_R7-0	Black level / adjustment. A black level of R signal after insert process is adjusted. (2's compl.)
7:0	BL_G7-0	Black level / adjustment. A black level of G signal after insert process is adjusted. (2's compl.)

BL_OFST1 (0003:07A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BL_B7	BL_B6	BL_B5	BL_B4	BL_B3	BL_B2	BL_B1	BL_B0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:8	BL_B7-0	Black level / adjustment. A black level of B signal after insert process is adjusted. (2's compl.)
7:0	RSV	Reserved Bit. Do not use.

MTX_GAIN0 (0003:07A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXRR7	MXRR6	MXRR5	MXRR4	MXRR3	MXRR2	MXRR1	MXRR0	MXGR7	MXGR6	MXGR5	MXGR4	MXGR3	MXGR2	MXGR1	MXGR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXRR7-0	Matrix coefficient setting. The MTX_RR value of RGB2RGB conversion matrix is set.
7:0	MXGR7-0	Matrix coefficient setting. The MTX_GR value of RGB2RGB conversion matrix is set.

$$\begin{bmatrix} R_{out} \\ G_{out} \\ B_{out} \end{bmatrix} = \begin{bmatrix} MTX_RR & MTX_GR & MTX_BR \\ MTX_RG & MTX_GG & MTX_BG \\ MTX_RB & MTX_GB & MTX_BB \end{bmatrix} \begin{bmatrix} R_{in} \\ G_{in} \\ B_{in} \end{bmatrix} + \begin{bmatrix} R_offset \\ G_offset \\ B_offset \end{bmatrix}$$

MXxx7 - MXxx0	Gain
1111 1111	X3.984375 (3 + 63 / 64)
0100 0000	x1
0010 0000	x0.5
0000 0001	x0.015625 (1/64)
0000 0000	x0

MTX_GAIN1 (0003:07AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXBR7	MXBR6	MXBR5	MXBR4	MXBR3	MXBR2	MXBR1	MXBR0	MXRG7	MXRG6	MXRG5	MXRG4	MXRG3	MXRG2	MXRG1	MXRG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15:8	MXBR7-0	Matrix coefficient setting. The MTX_BR value of RGB2RGB conversion matrix is set.
7:0	MXRG7-0	Matrix coefficient setting. The MTX_RG value of RGB2RGB conversion matrix is set.

MTX_GAIN2 (0003:07AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXGG7	MXGG6	MXGG5	MXGG4	MXGG3	MXGG2	MXGG1	MXGG0	MXBG7	MXBG6	MXBG5	MXBG4	MXBG3	MXBG2	MXBG1	MXBG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXGG7-0	Matrix coefficient setting. The MTX_GG value of RGB2RGB conversion matrix is set.
7:0	MXBG7-0	Matrix coefficient setting. The MTX_BG value of RGB2RGB conversion matrix is set.

MTX_GAIN3 (0003:07AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXRB7	MXRB6	MXRB5	MXRB4	MXRB3	MXRB2	MXRB1	MXRB0	MXGB7	MXGB6	MXGB5	MXGB4	MXGB3	MXGB2	MXGB1	MXGB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MXRB7-0	Matrix coefficient setting. The MTX_RB value of RGB2RGB conversion matrix is set.
7:0	MXGB7-0	Matrix coefficient setting. The MTX_GB value of RGB2RGB conversion matrix is set.

MTX_GAIN4 (0003:07B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MXBB7	MXBB6	MXBB5	MXBB4	MXBB3	MXBB2	MXBB1	MXBB0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:8	MXBB7-0	Matrix coefficient setting. The MTX_BB value of RGB2RGB conversion matrix is set.
7:0	RSV	Reserved Bit. Do not use.

MTX_OFST0 (0003:07B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	MXR9	MXR8	MXR7	MXR6	MXR5	MXR4	MXR3	MXR2	MXR1	MXR0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved Bit. Do not use.
9:0	MXR9-0	Hosu set value setting. The R_offset value of the RGB2RGB conversion matrix is set.

MTX_OFST1 (0003:07B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	MXG9	MXG8	MXG7	MXG6	MXG5	MXG4	MXG3	MXG2	MXG1	MXG0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved Bit. Do not use.
7:0	MXG9-0	Hosu set value setting. The G_offset value of the RGB2RGB conversion matrix is set.

MTX_OFST2 (0003:07B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	MXB9	MXB8	MXB7	MXB6	MXB5	MXB4	MXB3	MXB2	MXB1	MXB0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved Bit. Do not use.
7:0	MXB9-0	Hosu set value setting. The B_offset value of the RGB2RGB conversion matrix is set.

BYP_GAM (0003:07B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TBSEL	BYPR	BYPG	BYPB
R/W	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0

Bits	Register Name	Description
15:4	RSV	Reserved Bit. Do not use.
3	TBSEL	Gamma table selection. Select the gamma table. 0: Table 0 1: Table 1
2	BYP_R	Gamma table by-pass. In R signal gamma table is not used and it is bypassed. 0: Use Table 1: By-pass Table
1	BYP_G	Gamma table by-pass. In G signal gamma table is not used and it is bypassed. ガン 0: Use Table 1: By-pass Table
0	BYP_B	Gamma table by-pass. In B signal gamma table is not used and it is bypassed. 0: Use Table 1: By-pass Table

CSC0 (0003:07BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CRY7	CRY6	CRY5	CRY4	CRY3	CRY2	CRY1	CRY0	CGY7	CGY6	CGY5	CGY4	CGY3	CGY2	CGY1	CGY0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CRY7-0	Matrix coefficient setting. The CSC_RY value of RGB2RGB conversion matrix is set.
7:0	CGY7-0	Matrix coefficient setting. The CSC_GY value of RGB2RGB conversion matrix is set.

$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} CSC_RY & CSC_GY & CSC_BY \\ CSC_RCB & CSC_GCB & CSC_BCB \\ CSC_RCR & CSC_GCR & CSC_BCR \end{bmatrix} \begin{bmatrix} Rin \\ Gin \\ Bin \end{bmatrix}$$

CSC1 (0003:07BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CBY7	CBY6	CBY5	CBY4	CBY3	CBY2	CBY1	CBY0	CRCB7	CRCB6	CRCB5	CRCB4	CRCB3	CRCB2	CRCB1	CRCB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CBY7-0	Matrix coefficient setting. The CSC_BY value of RGB2RGB conversion matrix is set.
7:0	CRCB7-0	Matrix coefficient setting. The CSC_RCB value of RGB2RGB conversion matrix is set.

CSC2 (0003:07BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CGCB7	CGCB6	CGCB5	CGCB4	CGCB3	CGCB2	CGCB1	CGCB0	CBCB7	CBCB6	CBCB5	CBCB4	CBCB3	CBCB2	CBCB1	CBCB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CGCB7-0	Matrix coefficient setting. The CSC_GCB value of RGB2RGB conversion matrix is set.
7:0	CBCB7-0	Matrix coefficient setting. The CSC_BCB value of RGB2RGB conversion matrix is set.

CSC3 (0003:07C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CRCR7	CRCR6	CRCR5	CRCR4	CRCR3	CRCR2	CRCR1	CRCR0	CGCR7	CGCR6	CGCR5	CGCR4	CGCR3	CGCR2	CGCR1	CGCR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CRCR7-0	Matrix coefficient setting. The CSC_RCR value of RGB2RGB conversion matrix is set.
7:0	CGCR7-0	Matrix coefficient setting. The CSC_GCR value of RGB2RGB conversion matrix is set.

CSC4 (0003:07C2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CBCR7	CBCR6	CBCR5	CBCR4	CBCR3	CBCR2	CBCR1	CBCR0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15:8	CBCR7-0	Matrix coefficient setting. The CSC_GCR value of RGB2RGB conversion matrix is set.
7:0	RSV	Reserved Bit. Do not use.

Y_OFST (0003:07C4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	OTY9	OTY8	OTY7	OTY6	OTY5	OTY4	OTY3	OTY2	OTY1	OTY0
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:10	RSV	Reserved Bit. Do not use.
9:0	OTY7-0	Offset Y. After YCbCr conversion, the Y signal Offset is set.

C_OFST (0003:07C6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OTCB7	OTCB6	OTCB5	OTCB4	OTCB3	OTCB2	OTCB1	OTCB0	OTCR7	OTCR6	OTCR5	OTCR4	OTCR3	OTCR2	OTCR1	OTCR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	OTCB7-0	Offset CB. After YCbCr conversion, the Cb signal Offset is set.
7:0	OTCR7-0	Offset CR. After YCbCr conversion, the Cr signal Offset is set.

CNT_BRT (0003:07C8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CNTR7	CNTR6	CNTR5	CNTR4	CNTR3	CNTR2	CNTR1	CNTR0	BRT7	BRT6	BRT5	BRT4	BRT3	BRT2	BRT1	BRT0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	CBTR7-0	Contrast. After offset adjustments, the contrast of Y signal is set.(x0~x15.9375)
7:0	BRT7-0	Bright. After contrast adjustment, bright of Y signal is set.

C_SUP0 (0003:07CA) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SUPEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved Bit. Do not use.

Bits	Register Name	Description
0	SUPEN	Color signal pseudocolor suppression. 0: Off 1: On

C_SUP1 (0003:07CC) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SUPTH7	SUPTH6	SUPTH5	SUPTH4	SUPTH3	SUPTH2	SUPTH1	SUPTH0	SUPG7	SUPG6	SUPG5	SUPG4	SUPG3	SUPG2	SUPG1	SUPG0	SUPTH7
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	SUPTH7-0	Various signal pseudocolor suppression threshold. Set the CSUP_TH.
7:0	SUPG7-0	Various signal pseudocolor suppression gain. Set the CSUP_G.

SETUP_Y (0003:07CE) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MAXY7	MAXY6	MAXY5	MAXY4	MAXY3	MAXY2	MAXY1	MAXY0	MINY7	MINY6	MINY5	MINY4	MINY3	MINY2	MINY1	MINY0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MAXY7-0	Y signal maximum value.Data more than set value is clipped to the set value.
7:0	MINY7-0	Y signal minimum value.Data less than set value is clipped to set value.

SETUP_C (0003:07D0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MAXC7	MAXC6	MAXC5	MAXC4	MAXC3	MAXC2	MAXC1	MAXC0	MINC7	MINC6	MINC5	MINC4	MINC3	MINC2	MINC1	MINC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	MAXC7-0	C Signal maximum value. Data more than set value is clipped to the set value.
7:0	MINC7-0	Csignal minimum value. Data less than set value is clipped to the set value.

